

Full-universe evaluation — $t = 2015$ (all $M[t]=0$ pairs, no sampling)

Method	PR-AUC	AUROC	F1	NDCG@20	Prec@20	CWR
RCA Persistence	0.245	0.652	0.196	0.133	0.137	0.336
Density	0.055	0.778	0.109	0.140	0.127	0.886
ECI	0.017	0.482	0.035	—	—	0.449
KNN (LLM embeddings)	0.035	0.646	0.074	0.083	0.076	0.682
XGBoost (unoptimized)	0.203	0.896	0.298	0.319	0.284	0.952
XGBoost (optimized)	0.228	0.902	0.313	0.358	0.324	0.958
GNN-4F	0.072	0.818	0.137	0.107	0.101	0.914
GNN-11F	0.081	0.829	0.149	0.131	0.117	0.923
GNN-11F+LLM	0.083	0.830	0.151	0.156	0.140	0.925
GNN-11F-LLM + RCA-skip (unoptimized)	0.198	0.893	0.301	0.298	0.274	0.945
GNN-11F-LLM + RCA-skip (optimized)	0.202	0.895	0.299	0.294	0.268	0.952

Bold row = best method overall (wins the most metrics: XGBoost (optimized)). ECI has no within-country ranking signal, so NDCG@20 / Prec@20 are marked — rather than reported off an arbitrary tie-break.